MIL-P-43112B

25 March 1970
SUPERSEDING
MIL-P-43112A
29 November 1967

MILITARY SPECIFICATION

PAPER, WRAPPING, MEAT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type of meat wrapping paper (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-P-268	-	Paper, Kraft, Untreated, Wrapping
MMM-A-260	-	Adhesive, Water-Resistant, (For Sealing Waterproofed Paper)
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

FSC 8135

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(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TECNHICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

Testing Methods, Recommended Practices, Specifications of the Technical Association of the Pulp and Paper Industry.

T 401 m-60	-	Fiber Analysis of Paper and Paperboard.
T 402 m-49	-	Conditioning Paper and Paperboard for Testing.
T 403 ts-63	-	Bursting Strength of Paper.
T 410 os-68	-	Weight Per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 414 ts-65	-	Internal Tearing Resistance of Paper.
T 477 m-47	-	Blocking Resistance of Paper and Flexible Material.
T 435 su-68	-	Hydrogen Ion Concentration (pH) of Paper Extracts - Hot Extraction Method.

(Copies of TAPPI testing methods may be obtained from the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, New York 10017).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street N.W., Washington, D.C. 20036.)

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill. 60606.)

3. REQUIREMENTS

3.1 Materials.- Meat wrapping paper shall be made from chemical wood pulp and any other nontoxic ingredients necessary to conform to the requirements of this specification.

3.1.1 Toxicity.- Meat wrapping paper shall contain no materials which are injurious to human health within the meaning of all provisions of the Federal Food, Drug and Cosmetic Act. The supplier shall furnish the contracting officer a statement as to the chemical composition of all additives used.

3.1.2 Physical properties.- The paper shall conform to the physical requirements in table I when tested as specified in 4.3.1.

TABLE I.- Physical requirements

Basis weight, 24 x 36-500 sheets	Bursting strength, minimum average	Tear strength, minimum average, machine direction
Pounds	Points	Grams
40+2	22.0	50

3.1.3 Color.- The color of meat wrapping paper shall be the natural color of its components, tan, beige, or peach, as specified (see 6.2).

3.2 Water Resistance.- The paper shall have a minimum water drop retention of 120 minutes, to the extent that the droplet is not completely absorbed into the paper when tested on both felt side and wire side as specified in 4.4.3.

3.3 Blocking.- The paper shall not adhere to itself when tested as specified in table II.

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3.4 Discoloration.- The meat wrapping paper shall contain no component that will cause discoloration of the meat wrapped therein when tested as specified in table II.

3.5 Color bleeding.- Any dyes that may be used in the production of meat wrapping paper shall be set to the extent that no color bleeding shall be evident when tested as specified in 4.4.2.

3.6 Odor.- There shall be no undesirable odor when the paper is tested as specified in 4.4.1.

* 3.7 Hydrogen ion concentration.- The ph value of the meat wrapping paper shall be no less than 4.5 and no more than 7.2 when tested as specified in table II.

3.8 Form.- Meat wrapping paper shall be furnished in rolls of the width specified in the contract or order, with a tolerance of ± 1/8 inch (see 6.2).

3.8.1 Roll Construction.- The paper web shall be evenly and firmly wound on a fiberboard core having an inside diameter of 3 inches ± 1/8 inch. The length of the core shall be the specified width of the paper with a tolerance of ± 1/8 inch. No wrinkles, folds, or sharp creases shall occur within the roll, and the edge of the paper shall not telescope beyond either edge of the core by more than 1/4 inch. The end of the paper shall be securely fastened on the outer ply of the roll to prevent unwinding.

3.8.2 Roll diameter.- The diameter of the finished roll of paper shall be 9 inches, with tolerances of - 1/8 inch and ± 1/4 inch. The average diameter of all rolls tested shall be not less than 9 inches.

3.8.3 Splices.- Splices shall be well secured so that no breakage occurs when the rolls are unwound. There shall be no more than two splices per roll, and the minimum length of paper between splices shall be 20 yards.

3.8.4 Plugs.- The finished rolls shall contain snug-fitting plugs in the openings at each end of the roll. The plugs shall have a center hole 1 inch in diameter with a tolerance of ± 1/8 inch.

3.9 Workmanship.- The paper shall be free from dirt, spots, slits, scuff marks, cuts, wrinkles, holes, and soft spots across the face of the roll. The rolls shall have even, clean cut edges, free from lint and dust.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

4.2.1 Inspection of materials and components.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents. The supplier shall furnish the contracting officer with a statement of the chemical nature of the additives used, if applicable.

4.2.2 Inspection of the end item.-

4.2.2.1 Examination of the end item.- The end item shall be examined for the defects set forth in the applicable subparagraphs, at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.2.1.6. A random sample shall be drawn from each lot of material for each color for examination of visual and dimensional characteristics. The lot for determining the sample size shall be expressed in units of rolls of a specified color for examinations in 4.2.2.1.1, 4.2.2.1.2 and 4.2.2.1.3

4.2.2.1.1 Examination of the end item for defects in appearance and general workmanship.- For examination of defects within rolls, the sample unit shall be 12 consecutive yards, full width of the roll. No more than three sample units shall be selected from any one roll. Both sides of the material shall be examined.

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Examine	Defect
Appearance	Not color specified. Color not uniform on both sides.
Workmanship	Surfaces not clean; presence of any dust, dirt, spots, lint,or other foreign matter. Any cut, slit, hole, wrinkle, scuff mark, or noticeable soft spot across the face of the roll.

4.2.2.1.2 Examination of the end item for dimensional defects.- The sample unit for this examination shall be one roll.

Examine	Defect
Width	Varies from specified width by more than 1/8 inch.
Diameter	Less than 8-7/8 inches. More than 9-1/4 inches. Average of all rolls tested less than 9 inches.
Core	Inside diameter less than 2-7/8 or more than 3-1/8 inches. Length varies from specified dimension by more than 1/8 inch.
Plug	Center hole diameter less than 7/8 inch or more than 1-1/8 inch.

4.2.2.1.3 Examination of the end item for defects in construction.- The sample unit for this examination shall be one roll.

Examine	Defect
Assembly of roll	Not suitably restrained to prevent unwinding. Paper telescopes on roll more than 1/4 inch. Core not rigid; broken, collapsed, crushed or otherwise mutilated. Plugs missing; plugs do not fit snugly.

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Examine	Defect
Assembly of roll (cont'd)	Edges not clean cut; nicked; presence of lint or dust.
Unwinding of roll	Roll wound unevenly causing wrinkles, sharp creases or folds within roll. More than 2 splices in any roll; less than twenty lineal yards between splices. Splices not secure; come apart during unwinding.

4.2.2.1.4 Examination for packing and marking.- An examination shall be made to determine that packing and marking requirements of section 5 are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one wrapped roll fully prepared for delivery. Wrapped rolls, fully prepared for delivery, shall be examined for defects of closure. The lot shall be the number of wrapped rolls in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application, incomplete.
Materials	Any component missing. Any component damaged.
Workmanship	Inadequate application of components, such as improper taping.

4.2.2.1.5 Examination for palletization.- An examination shall be made to determine that the palletization requirements of section 5 are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

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Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Wood cap not used over and under the load. Double tray cap (when applicable) not used. Load not bonded with required straps as specified. Pallet damaged. Straps not properly tensioned to immobilize load.
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application, incomplete.

4.2.2.1.6 Inspection levels and acceptable quality levels (AQLs) for examinations.- The inspection levels for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQLs
4.2.2.1	S-4	2.5
4.2.2.1.2	S-2	2.5
4.2.2.1.3	S-1	2.5

4.3 Examination of the end item.-

4.3.1 Testing of the end item.- The end item shall be tested for the applicable characteristics as indicated in table II for each lot of the same color offered for inspection at one time. The sample size shall be one piece of 48 inch length, full width of the roll. No more than one sample unit shall be removed from any one roll, and the outer layer shall be removed from the roll before taking the sample. There shall be no failure of any sample unit or composite sample to meet the specified requirements. All test reports shall contain the individual values utilized in expressing the final result. The inspection sampling shall be as follows:

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Sampling plan

Lot size (rolls)	Sample size (No. of sampling units)
800 or less	2
801 to and including 22,000	3
22,001 or more	5

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INSTRUCTIONS FOR TESTING OF THE END ITEM

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Atmospheric conditions		T402m-49 1/			All testing				
Fiber identification	3.1	T401m-60 1/			2 on composite 2/		5 percent		
Toxicity	3.1.1	3/	X		1 (average of 5 specimens)	X	0.1 pound		
Basis weight, (24 x 36 - 500)	3.1.2	T410os68							
Bursting strength	3.1.2	T403ts63 1/	X		Average of 10 (1 on each side of 5 specimens)		point		
Tear strength, machine direction	3.1.2	T414ts65 1/	X		Average of 5		gram		
Water resistance	3.2	443	X		1	X			
Felt side									
Wire side									
Blocking	3.3	T477m-47 4/	X		1	X			
Discoloration	3.4	5/							
Color bleeding (where applicable)	3.5	4.4.2	X		1	X			
Odor	3.6	4.4.1	X		1	X			
Hydrogen ion concentration	3.7	T435su68 1/	X		1	X	0.1 ph		

1/ Refers to TAPPI standard test number.
2/ Composite shall consist of small, approximately equal portions taken from all sample units.
3/ Supplier's Certificate of Compliance regarding conformance to Food, Drug, and Cosmetic Act shall satisfy requirement.
4/ (TAPPI Std Number) Tests shall be run at 140°F. and 44 percent R.H.
5/ Supplier's Certificate of Compliance shall state that the paper does not discolor the meat.

AMXRE Form 598
8 February 1965

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4.4 Test procedures.-

4.4.1 Odor test.- A 12-by-12 inch specimen of meat wrapping paper shall be cut into small pieces and submerged in a clean beaker containing distilled water at room temperature. After 2 hours, the odor shall be compared with that of another beaker containing distilled water only to determine compliance with 3.6.

4.4.2 Color bleeding test.- Remove the paper from the beaker used in 4.4.1 and compare the color of the water with that in the beaker which contained distilled water only to determine compliance with 3.5.

4.4.3 Water resistance test.- Place two 4-by-4 inch specimens of the paper on a clean, dry surface; one specimen with the wire side up, the other with the felt side up. Drop approximately 0.1 milliliter of distilled water at 73° ± 3.5°F on the surface of each specimen from a height not to exceed 3 inches, and immediately cover each drop with a watch glass. The paper shall then be examined to determine compliance with 3.2.

5. PREPARATION FOR DELIVERY.

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.- Each roll of paper shall be wrapped overall with waterproofed, kraft wrapping paper conforming to PPP-B-1055, except that when class C-1 is used, two thicknesses shall be required. The wrapper shall be closed at the ends by means of inside and outside headers of the same material as the wrapper. Seams and headers shall be sealed with a water-resistant adhesive conforming to type I or II, grade A, class 2 or 3 of MMM-A-260, using sufficient adhesive to effect a watertight seal.

5.1.2 Level B.- Each roll of paper shall be wrapped overall with minimum 80-pound basis weight kraft wrapping paper conforming to grade A or B of UU-P-268. The wrapper shall be closed at the ends by means of inside and outside headers of the same material as the wrapper. Seams and headers shall be securely sealed with an adhesive commercially used for this purpose or by minimum 2-inch width gummed paper tape conforming to type III, grade C of PPP-T-45.

5.1.3 Level C.- Rolls of papers of one description only shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Shipments shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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* 5.2 Palletization.- Unless otherwise specified (see 6.2), rolls of paper, packed as specified, shall be palletized in accordance with load type XIII of MIL-STD-147. Rolls of paper shall be stacked vertically in single or multiple courses (as applicable) with a wood cap over and under the load and a double tray cap (as applicable) between courses. A strap shall be positioned around the load, outside of and at the vertical center of the rolls of paper in each course, and shall be the first strap applied to the load. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L.

5.3 Marking.- In addition to any special marking required by the contract or order, wrapped rolls and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use.- Meat wrapping paper is to be used for the wrapping of meat for delivery from commissaries or stores, and for short periods of refrigerated storage.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color (see 3.1.3).
(c) Width of rolls (see 3.8).
(d) Inspection responsibility, methods, and tests (see Section 4).
(e) Selection of the applicable level of packing required (see 5.1).
(f) When palletization is not required (see 5.2).

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the prior issue.

Custodians:

Army - GL
Navy - MC
Air Force - 69

Preparing activity:

Army - GL

Project No. 8135-0307

Review activities:

Army - MD
Navy - MC, SA
Air Force - 69, 84

User activity:

Army - SM

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-P-43112B Paper, Wrapping, Meat

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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